

On Total Vertex Cover Problem in Subcubic Graphs

Sheung-Hung Poon^{*1} and Wei-Lin Wang²

¹School of Computing and Informatics, Universiti Teknologi Brunei,
Gadong, Brunei Darussalam

²Institute of Information Systems and Applications, National Tsing Hua University,
Hsinchu, Taiwan

Abstract

A total vertex cover is a vertex cover whose induced subgraph consists of a set of connected components, each of which contains at least two vertices. The total vertex cover (TVC) problem asks for the total vertex cover with minimum cardinality. In this paper, we first show that the TVC problem is NP-complete for connected subcubic grid graphs of arbitrary large girth. Next, we show that the TVC problem is NP-complete for 3-connected cubic planar graphs. Moreover, we show that the TVC problem is APX-complete for connected subcubic graphs of arbitrary large girth.

1 Introduction

A vertex cover C is a subset of vertices of a graph such that every edge of the graph is incident to at least one vertex in C . A vertex cover C is called *total*, if it is a vertex cover of a graph such that, each vertex is adjacent to at least one other vertex in C . The *vertex cover (VC) problem* and the *total vertex cover (TVC) problem*, each asks for a corresponding set with minimum cardinality. These formulated problems appear in various applications, such as placing minimum number of ATMs or security cameras at the corners of streets [6].

Next, we define several technical terms used in this paper. A *cubic graph* (also called a *3-regular graph*) is a graph where every vertex has degree three. A *subcubic graph* is a graph where every vertex has degree at most three. A *grid graph* is an induced subgraph of a set of vertices from two dimensional grid. A *k-connected graph* is a graph which does not contain a set of $k-1$ vertices whose

removal disconnects the graph. A ρ -*path* is a path of ρ vertices. A ρ -*cycle* is a cycle of ρ vertices. The *girth* of a graph is the shortest cycle length in the graph. The girth is considered to be infinity if the graph has no cycle. A connected component of a graph is a maximal connected subgraph of the graph. A *family* $\{a_i\}_{i \in I}$ is a set of elements, such that an element a_i in the family is associated with an index $i \in I$.

The VC problem is an important classical graph optimization problem. The complexity of VC problem has been investigated in various works. Uehara [7] showed that the VC problem is NP-complete for 3-connected cubic planar graphs. In this paper, we show that the TVC problem is also NP-complete for 3-connected cubic planar graphs.

The TVC problem is a close variation of the VC problem, and is investigated in depth in this paper. First, we present the related work on NP-completeness results for the TVC problem. The problem is first introduced by Fernau and Manlove [4]. They showed that the TVC problem is NP-complete for planar subcubic graphs. In this paper, we make a step forward to show that TVC problem is NP-complete for connected planar subcubic grid graphs of arbitrary large girth.

There are also some existing work on approximability results for the TVC problem. Fernau [4] showed that the TVC problem is approximable within two for general graphs. This implies that the TVC problem is in APX for general graphs. In this paper, we further show that the TVC problem is APX-complete for subcubic graphs of arbitrary large girth.

^{*}Corresponding author: sheung.hung.poon@gmail.com

2 NP-completeness for subcubic grid graphs of arbitrary girth

This section starts by showing that the TVC problem is NP-complete for connected subcubic grid graphs of arbitrary large girth. The NP-hardness proof is via reduction from the rectilinear planar monotone 3-SAT problem, which is known to be NP-complete [3], and uses very intricate gadgets in the reduction.

Theorem 1. *The TVC problem is NP-complete for connected subcubic grid graphs of arbitrary large girth.*

Proof. We first show that the TVC problem is in NP. The decision version of the TVC problem asks if there exists a total vertex cover of size of at most k , for a given fixed number k . We solve the decision problem through a polynomial number of guessing of the total vertex cover of size of at most k . In each guessing, each vertex set in the total vertex cover has its incident edges removed. After the edge removal, if there is no edge left, output “yes”; otherwise, output “no”. Thus, the TVC problem is in NP. To show that the TVC problem is NP-hard, we reduce from the rectilinear planar monotone 3-SAT problem [3].

The input instance of the rectilinear planar monotone 3-SAT problem is a rectilinear representation drawing of a Boolean formula Φ . The Boolean formula $\Phi = c_0 \wedge \dots \wedge c_{m-1}$ is a conjunction of m clauses. Each clause $c_h = l'_{\alpha,h} \vee l'_{\beta,h} \vee l'_{\gamma,h}$ of the Boolean formula Φ is a disjunction of three literals. The three literals of a clause are either any three variables in the family $\{x_i\}_{i=0}^{n-1}$, called *positive literals*, or the negation of the three variables, called *negative literals*. The rectilinear representation drawing consists of:

- (a) A variable gadget X'_i for each variable x_i , represented by horizontal segments lying on the x -axis.
- (b) A clause gadget D'_h for each clause c_h , represented by horizontal segments lying on the upper side the x -axis if it consists of only positive true literals, or lying on the lower side of the x -axis if it consists of only negative literals.
- (c) A literal gadget $L'_{\alpha,h}$, $L'_{\beta,h}$, or $L'_{\gamma,h}$ for each of the three literals of clause c_h , represented by the vertical segment which links the clause gadget D'_h and the variable gadget X'_j together.

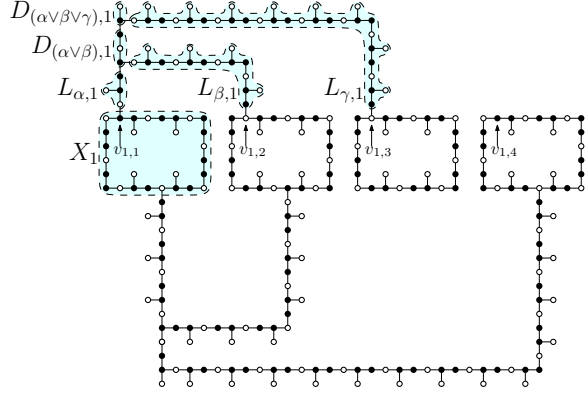


Figure 1: A constructed graph G for the Boolean expression $\Phi = (x_1 \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_4})$ with a truth assignment $x_1 = \text{TRUE}$, $x_2 = \text{FALSE}$, $x_3 = \text{FALSE}$, and $x_4 = \text{TRUE}$. We let $p = 2$ here; thus, the girth of the graph is $6(m+p) = 24$. Solid bullets represent the 138 vertices in a minimum total vertex cover of G .

The outcome of the rectilinear planar monotone 3-SAT is “yes” if the Boolean formula is satisfiable, and “no” otherwise. In the following, we proceed to construct the TVC problem.

The input instance of the TVC problem is a connected subcubic grid graph G of arbitrary large girth g . The graph G is constructed according to the rectilinear representation drawing of the Boolean formula Φ . We modify the Boolean formula Φ , so that each of its clause $c = l'_{\alpha,h} \vee l'_{\beta,h} \vee l'_{\gamma,h} = \neg(l_{\alpha,h} \wedge l_{\beta,h} \wedge l_{\gamma,h})$, where $l_{\alpha,h} = \neg l'_{\alpha,h}$, $l_{\beta,h} = \neg l'_{\beta,h}$, $l_{\gamma,h} = \neg l'_{\gamma,h}$. However, the rectilinear representation drawing of the Boolean formula Φ remains the same. Graph G contains the following families of gadgets:

- (1) The family of variable gadgets $\{X'_i\}_{i=0}^{n-1}$. For any variable gadget X'_i , its vertex set is $\{v_{j,i}\}_{j=1}^{6(m+p)} \cup \{d_{3j-1,i}\}_{j=1}^m \cup \{d_{3j-1,i}\}_{j=m+p+1}^{2m+p}$, for a specified number $p \in \mathbb{N} \setminus \{1\}$. (the second index of $v_{j,i}$ and $d_{3j+2,i}$ might be ignored for brevity). These vertices induce a $6(m+p)$ -cycle $v_1 \dots v_{6(m+p)}v_1$ and two families of edges $\{\{v_{3j-1}, d_{3j-1}\}\}_{j=1}^m$ and $\{\{v_{3j-1}, d_{3j-1}\}\}_{j=m+p+1}^{2m+p}$. Two subgraphs of the gadget, $v_{6(m+p)}v_1 \dots v_{3m}v_{3m+1}$ and $v_{3(m+p)} \dots v_{3(2m+p)+1}$, are $(3m+2)$ -paths parallel to the x -axis; the other two subgraphs of the gadget, $v_{3m+1} \dots v_{3(m+p)}$ and $v_{3(2m+p)+1} \dots v_{6(m+p)}$, are $3p$ -paths parallel to the y -axis. For a pair of variable gadgets in the family $\{(X'_i, X'_{i+1})\}_{i=0}^{n-2}$, they have the horizontal distance 2.

(2) The literal gadgets:

- The literal gadget $L_{\alpha,h}$ for the first literal $l_{\alpha,h}$ of clause c_h . The vertex set of $L_{\alpha,h}$ is $\{u_{j,\alpha,h}\}_{j=1}^{3(2h+1)} \cup \{d_{3j-1,\alpha,h}\}_{j=1}^{2h+1}$ (the second and third indices of $u_{j,\beta,h}$ and $d_{3j-1,\alpha,h}$ might be ignored for brevity). These vertices induce a $(3(2h+1))$ -path $u_1 \dots u_{3(2h+1)}$ parallel to y -axis and the family of edges $\{\{v_{3j-1}, d_{3j-1}\}\}_{j=1}^{2h+1}$.
- The literal gadget $L_{\beta,h}$ for the second literal $l_{\beta,h}$ of clause c_h , which is similar to $L_{\alpha,h}$.
- The literal gadget $L_{\gamma,h}$ for the third literal $l_{\gamma,h}$ of clause c_h , which is similar to $L_{\alpha,h}$.

(3) The clause gadgets:

- The clause gadget $D_{(\alpha\vee\beta),h}$ for the clause $c_{(\alpha\vee\beta),h} = l_{\alpha,h} \vee l_{\beta,h}$. The vertex set of $D_{(\alpha\vee\beta),h}$ is $\{u_{j,(\alpha\vee\beta),h}\}_{j=1}^3 \cup \{d_{(\alpha\vee\beta),h}\}$. (the last two indices of $u_{j,(\alpha\vee\beta),h}$ and $d_{(\alpha\vee\beta),h}$ might be ignored for brevity). These vertices induce a 3-path $u_1 u_2 u_3$ parallel to the y -axis and the edge $\{u_2, d\}$.
- The clause gadget $D_{(\alpha\vee\beta\vee\gamma),h}$ for the clause $c_{(\alpha\vee\beta\vee\gamma),h} = c_{(\alpha\vee\beta),h} \vee l_{\gamma,h} = l_{\alpha,h} \vee l_{\beta,h} \vee l_{\gamma,h}$. The vertex set of $D_{(\alpha\vee\beta\vee\gamma),h}$ is the family $\{u_{j,(\alpha\vee\beta\vee\gamma),h}\}_{j=1}^2$ of vertices (the second and third indices of $u_{j,(\alpha\vee\beta\vee\gamma),h}$ might be ignored for brevity). These vertices induce an edge $\{u_1, u_2\}$ parallel to the y -axis.

The gadgets are linked together as specified in the following. $D_{(\alpha\vee\beta\vee\gamma),h}$ links to $D_{(\alpha\vee\beta),h}$ and $L_{\gamma,h}$ through the edges $\{u_{1,(\alpha\vee\beta\vee\gamma),h}, u_{3,(\alpha\vee\beta),h}\}$ and $\{u_{1,(\alpha\vee\beta\vee\gamma),h}, u_{3(2h+2+(\gamma-\alpha)(m+1)),\gamma,h}\}$. $D_{(\alpha\vee\beta),h}$ links to $L_{\alpha,h}$ and $L_{\beta,h}$ through the edges $\{u_{1,(\alpha\vee\beta),h}, u_{3(2h+1),\alpha,h}\}$ and $\{u_{1,(\alpha\vee\beta),h}, u_{3(2h+1+(\beta-\alpha)(m+1)),\beta,h}\}$. If literal $l_{\alpha,h} = x_\alpha$, then $L_{1,\alpha,h}$ links to X_α through the edge $\{u_{1,\alpha,h}, v_{(1+3h),\alpha}\}$; otherwise, if literal $l_{\alpha,h} = \overline{x_\alpha}$, then $L_{1,\alpha,h}$ links to X_α through the edge $\{u_{1,\alpha,h}, v_{(3(2m+p)-3h),\alpha}\}$. In similar manners, $L_{\beta,h}$ links to X_β , and $L_{\gamma,h}$ links to X_γ .

The girth g of the constructed connected sub-cubic grid graph G is $6(m+p)$, for a specified number $p \in \mathbb{N} \setminus \{1\}$.

The outcome of the TVC problem is a minimum total vertex cover C , which is reduced from a truth assignment on the variables of Boolean formula

Φ . Here we only show an outline of the complex construction of the minimum total vertex cover.

- (i) We first consider the minimum total vertex cover C of graph G on variable gadgets. Every variable gadget $X_i = (V_i, E_i)$ has two alternative choices of $C \cap V_i$ of $4(m+p)$ vertices: One is chosen by us when variable $x_i = \text{TRUE}$. The other is chosen by us when variable $x_i = \text{FALSE}$.

- (ii) We then consider the minimum total vertex cover C of graph G on literal gadgets.

- Every literal gadget $L_{\alpha,h} = (V_{\alpha,h}, E_{\alpha,h})$ for the first literal $l_{\alpha,h}$ of clause c_h has two alternative categories of choices of $C \cap V_{\alpha,h}$ of $2(2h+1) = 2q_\alpha$ vertices. One is chosen by us when either the literal $l_{\alpha,h} = x_\alpha = \text{TRUE}$, or $l_{\alpha,h} = \overline{x_\alpha} = \text{TRUE}$. The other is chosen by us when either the literal $l_{\alpha,h} = x_\alpha = \text{FALSE}$, or $l_{\alpha,h} = \overline{x_\alpha} = \text{FALSE}$.
- Every literal gadget $L_{\beta,h} = (V_{\beta,h}, E_{\beta,h})$ for the second literal $l_{\beta,h}$ of clause c_h has two alternative categories of choices of $C \cap V_{\beta,h}$ of $2(2h+1+(\beta-\alpha)(m+1)) = 2q_\beta$ vertices. One is chosen by us when either the literal $l_{\beta,h} = x_\beta = \text{TRUE}$, or $l_{\beta,h} = \overline{x_\beta} = \text{TRUE}$. The other is chosen by us when either the literal $l_{\beta,h} = x_\beta = \text{FALSE}$, or $l_{\beta,h} = \overline{x_\beta} = \text{FALSE}$.
- Every literal gadget $L_{\gamma,h} = (V_{\gamma,h}, E_{\gamma,h})$ for the third literal $l_{\gamma,h}$ of clause c_h has two alternative categories of choices of $C \cap V_{\gamma,h}$ of $2(2h+2+(\gamma-\alpha)(m+1)) = 2q_\gamma$ vertices, similar to that of $L_{\beta,h}$.

- (iii) We then consider the minimum total vertex cover C of graph G on clause gadgets.

- Every clause gadget $D_{(\alpha\vee\beta),h} = (V_{(\alpha\vee\beta),h}, E_{(\alpha\vee\beta),h})$ for the clause $c_{(\alpha\vee\beta),h}$ has two alternative categories of choices of $C \cap D_{(\alpha\vee\beta),h}$ of 2 vertices. One is chosen by us when the clause $c_{(\alpha\vee\beta),h} = \text{TRUE}$. The other is chosen by us when the clause $c_{(\alpha\vee\beta),h} = \text{FALSE}$.
- Every clause gadget $D_{(\alpha\vee\beta\vee\gamma),h} = (V_{(\alpha\vee\beta\vee\gamma),h}, E_{(\alpha\vee\beta\vee\gamma),h})$ for the clause $c_{(\alpha\vee\beta\vee\gamma),h}$ has two alternative categories of choices of $C \cap D_{(\alpha\vee\beta\vee\gamma),h}$. One is chosen by us when the clause

$c_{(\alpha \vee \beta \vee \gamma),h} = \text{TRUE}$. The other is chosen by us when the clause $c_{(\alpha \vee \beta \vee \gamma),h} = \text{FALSE}$, i.e. the Boolean formula Φ is not satisfied.

Thus, the Boolean formula Φ is satisfied by a truth assignment if and only if the constructed graph G has a total vertex cover C of size at most $n \cdot 4(m + p) + \sum_{h=0}^{m-1} (2(h+1) + 2(2h+1 + (\beta - \alpha)(m+1)) + 2(2h+2 + (\gamma - \alpha)(m+1)) + 2 + 1) = 4n(m + p) + \sum_{h=0}^{m-1} (2(5h+4 + (\beta + \gamma - 2\alpha)(m+1)) + 3)$. This completes the proof. $\square$

3 NP-completeness for 3-connected cubic planar graphs

This section starts by showing that the TVC problem is NP-complete for 3-connected cubic planar graphs. The NP-hardness proof is via reduction from VC problem for any cubic planar graph, which is known to be NP-complete [5], and uses very intricate gadgets in the reduction.

Lemma 1. [7] *A cubic graph is 3-connected if and only if it is 3-edge-connected.*

To prove the NP-completeness for 3-connected cubic planar graphs, due to Lemma 1, we only need to prove the NP-completeness for 3-edge-connected cubic planar graphs, where a *k-edge-connected graph* is a graph that does not contain a set of $k - 1$ edges whose removal disconnects the graph.

Theorem 2. *The TVC problem is NP-complete for 3-connected cubic planar graphs.*

Proof. Clearly, this problem is in NP. To show that this problem is NP-hard, we show a polynomial-time reduction from a known NP-complete problem, the VC problem for any cubic planar graph G_1 [5]. The reduction has three steps. First, the VC problem for any cubic planar graph G_1 is reduced to the first TVC problem for a 2-edge-connected cubic planar graph G_2 . Second, the first TVC problem for a 2-edge-connected cubic planar graph G_2 is reduced to the second TVC problem for a 3-edge-connected cubic planar graph G_3 . Third, we use the lemma given by Uehara [7] to show that G_3 is a 3-edge-connected cubic planar graph if and only if G_3 is a 3-connected cubic planar graph.

We show the first step of the reduction.

We formulate the VC problem in the following. The input instance of the VC problem is a cubic

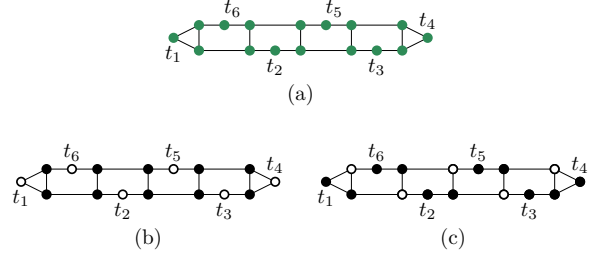


Figure 2: (a) The gadget H for the TVC problem, which is reduced from a degree-6 vertex v , and has the six neighbors of vertex v adjacent to the vertices t_1 to t_6 . (b) The minimum total vertex cover of the gadget H consisting of the 10 black vertices. (c) A total vertex cover of size of 11 on the gadget H represented by the black vertices.

planar graph G_1 . The outcome of the VC problem is the minimum vertex cover C_1 of graph G_1 .

We formulate the first TVC problem in the following.

The input instance of the first TVC problem is a 2-edge-connected cubic planar graph G_2 . G_2 is reduced from cubic planar graph G_1 . For each edge $e_1 = \{u, v\}$ of G_1 , we add a parallel edge $e'_1 = \{u, v\}$ to G_1 . Hence, each vertex of G_1 becomes a vertex of degree 6. Then we map each vertex of degree 6 to the gadget H in graph G_2 shown in Figure 2(a). We also map each of the 6 incident edges of the degree-6 vertex to an incident edge of each of the vertices t_1 to t_6 , respectively.

The outcome of the first TVC problem is a minimum total vertex cover C_2 of graph G_2 , which is reduced from the minimum vertex cover C_1 of graph G_1 . For every vertex of G_1 set in C_1 , it corresponds to the minimum total vertex cover of the gadget H shown in Figure 2(b), which has size of 10; for every vertex of G_1 set not in C_1 , it corresponds to the total vertex cover of size of 11 on the gadget H shown in Figure 2(c).

Then we show the reduction from VC problem to first TVC problem.

Lemma 2. *Cubic planar graph G_1 has a vertex cover C_1 of size k_1 if and only if 2-connected cubic planar graph G_2 has a total vertex cover C_2 of size $k_1 + 10n_1$, where n_1 is the number of vertices of graph G_1 .*

We omit the proof. This completes the first step of the reduction.

We show the second step of the reduction. The first TVC problem has been formulated in the first step. Now we formulate the second TVC problem

in the following.

The input instance of the second TVC problem is a 3-edge-connected cubic planar graph G_3 . G_3 is reduced from 2-connected cubic planar graph G_2 . For each edge $e_2 = \{u, v\}$ of G_2 , we add a parallel edge $e'_2 = \{u, v\}$ to G_2 . Hence, each vertex of G_2 becomes a vertex of degree 6. Then we map each vertex of degree 6 to the gadget H in graph G_3 shown in Figure 2(a). We also map each of the 6 incident edges of the degree-6 vertex to an incident edge of each of the vertices t_1 to t_6 , respectively.

The outcome of the second TVC problem is a minimum total vertex cover C_3 of graph G_3 , which is reduced from the minimum vertex cover C_2 of graph G_2 . For every vertex of G_2 set in C_2 , it corresponds to the minimum total vertex cover of the gadget H shown in Figure 2(b), which has size of 10; for every vertex of G_2 set not in C_2 , it corresponds to the total vertex cover of size of 11 on the gadget H shown in Figure 2(c).

Then using similar arguments as in our first step of reduction, we can show that 2-connected cubic planar graph G_2 has a vertex cover C_2 of size k_2 if and only if 3-connected cubic planar graph G_3 has a total vertex cover C_3 of size $k_2 + 10n_2$, where n_2 is the number of vertices of graph G_2 . This completes the second step of the reduction.

Lemma 1 says that G_3 is a 3-edge-connected cubic planar graph if and only if G_2 is a 3-connected cubic planar graph. Using this lemma, we complete the proof. $\square$

4 Approximation hardness for connected subcubic graphs of arbitrary girth

Fernau and Manlove [4] showed that the TVC problem for general graphs is approximable within two. Thus, the TVC problem for general graphs, including subcubic graphs, is in APX. In this section, we further prove that the TVC problem for subcubic graphs is in fact APX-complete.

To prove the APX-completeness of a problem, we present an approximation-preserving reduction, called *L-reduction* (“linear reduction”) [2], from the other APX-complete problem to this problem. An L-reduction preserves the approximability and the relative errors.

Then we prove that the TVC problem for connected graphs of subcubic graphs of arbitrary large girth is APX-complete.

We show an L-reduction from the VC problem for connected subcubic graphs, which is known

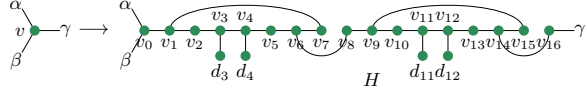


Figure 3: The function f of the L-reduction replaces each vertex v of graph G by the gadget H in graph G_2 of girth $g = 3k + 1 = 7$, when we let $k = 2$.

to be APX-complete [1], to the TVC problem for connected subcubic graphs of arbitrary large girth g . The instance of the VC problem is a connected subcubic graph $G_1 = (V_1, E_1)$. The instance of the TVC problem is a connected subcubic graph $G_2 = (V_2, E_2)$ of arbitrary large girth g . The function f of the L-reduction is from the set of every G_1 to the set of every G_2 . We then describe the function f in detail. We are given a graph $G_1 = (V_1, E_1)$. The function f then map each vertex $v \in V_1$ to a gadget $H = (V_H, E_H)$, where $V_H = \{v_i\}_{i=0}^{k(g+1)} \cup \{\{d_{3(j+1)+s+i(g+1)}\}_{s=0}^1\}_{j=0}^{k-1}\}_{i=0}^{k-1}$, and $E_H = \{v_{i(g+1)}v_{1+i(g+1)}, v_{(j \bmod g)+1+i(g+1)}v_{(j+1 \bmod g)+1+i(g+1)}, v_{-2+(i+1)(g+1)}v_{(i+1)(g+1)}, \{v_{3(j+1)+s+i(g+1)}d_{3(j+1)+s+i(g+1)}\}_{s=0}^1\}_{j=0}^{k-1}\}_{i=0}^{k-1}$, $g = 3k + 1$, $k \in \mathbb{N}$. Moreover, for each vertex $v \in V_1$, the function f maps the set of its existing incident edges $\{\alpha v, \beta v, \gamma v \mid N(v) = \{\alpha, \beta, \gamma\}\}$ to the set of incident edges of some vertices of gadget H : $\{\alpha v_0, \beta v_0, \gamma v_{k(g+1)} \mid v_0, v_{k(g+1)} \in V_H\}$. See Figure 3 for an example. The resulting graph is G_2 .

For the gadget H , the following lemma holds.

Lemma 3. *A total vertex cover of gadget H has size $2k^2 + k$, $k \in \mathbb{N}$, if and only if it does not include the vertex set $\{v_0, v_{k(g+1)}\}$. Moreover, a total vertex cover of gadget H has size of $2k^2 + k + 1$ if and only if it includes the vertex set $\{v_0, v_{k(g+1)}\}$.*

We omit the proof.

The solution of the TVC problem is a total vertex cover D on a connected subcubic graph $G_2 = (V_2, E_2)$ of arbitrary large girth g . The solution of the VC problem is a vertex cover C on a connected subcubic graph $G_1 = (V_1, E_1)$ of arbitrary large girth g . From the reduction, we have the function h from the set of every D to the set of every C . We define function h in detail in the following lemma.

Lemma 4. *There is a polynomial-time reduction h such that for each total vertex cover D in G_2 , if $|D \cap V_H| = 2k^2 + k + 1$, then we replace $D \cap V_H$ by a vertex in $h(D)$; otherwise, if $|D \cap V_H| = 2k^2 + k$,*

then we replace $D \cap V_H$ by a vertex in $(V_1 \setminus h(D))$. Moreover, $h(D)$ is a vertex cover in G_1 .

We omit the proof.

We also introduce the inverse function of the function h , that is the function h^{-1} from the set of every C to the set of every D . We describe function h^{-1} in detail in the following lemma.

Lemma 5. *The inverse h^{-1} of the polynomial-time reduction h satisfies the following condition. For each vertex cover C in G_1 , if v is in C , then we replace v by $(h^{-1}(C) \cap V_H)$ of size $2k^2 + k + 1$; otherwise, if v is in $(V_1 \setminus C)$, then we replace v by $(h^{-1}(C) \cap V_H)$ of size $2k^2 + k$. Moreover, $h^{-1}(C)$ is a total vertex cover in G_2 .*

We omit the proof.

Then by performing some calculations, we can derive that $\alpha = 8k^2 + 4k + 1$, and $\beta = 1$. Details are omitted here. Thus we obtain the L-reduction.

Hence, we have the following theorem.

Theorem 3. *The TVC problem is APX-complete for connected subcubic graphs of arbitrary large girth g .*

5 Conclusion & Discussion

We have shown the NP-completeness of the TVC problem in subcubic grid graphs of arbitrary large girth and in 3-connected cubic planar graphs, respectively. Moreover, we have also shown the APX-completeness of the TVC problem in subcubic graphs of arbitrary large girth.

The above NP-completeness results make us being curious about the complexity of the TVC problem on cubic grid graphs. As it is clear that cubic grid graphs in two dimensions do not exist, we consider the *three-dimensional cubic grid graphs*, each of which is defined to be an induced subgraph of a set of vertices from three dimensional grid, such that every graph vertex has degree exactly three. We leave open the interesting question whether the TVC problem in three-dimensional cubic grid graphs is NP-complete.

References

- [1] P. Alimonti and V. Kann. Some APX-completeness results for cubic graphs. *Theoretical Computer Science*, 237(1–2):123–134, 2000.
- [2] P. Crescenzi. A short guide to approximation preserving reductions. In *Proceedings of the 12th Annual IEEE Conference on Computational Complexity*, 262–273, 1997.
- [3] M. de Berg and A. Khosravi. Optimal binary space partitions in the plane. In *Proceedings of the 16th International Conference on Computing and Combinatorics*, 329–343, 2010.
- [4] H. Fernau and D. F. Manlove. Vertex and edge covers with clustering properties: complexity and algorithms. *Journal of Discrete Algorithms*, 7(2):149–167, 2009.
- [5] M. R. Garey and D. S. Johnson. *Computers and intractability: a guide to the theory of NP-completeness*. W. H. Freeman & Co., 1990.
- [6] Gurleen Singh, Geetanjali Sharma, and Prabhdeep Singh. A novel algorithm to solve vertex cover problem. *International Journal of Advanced Computational Engineering and Networking*, 24(3):225–236, 2014.
- [7] R. Uehara. NP-complete problems on a 3-connected cubic planar graph and their applications. *Technical Report TWCU-M-0004*, Tokyo Woman's Christian University, 1996.